PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
SPECIAL SCHOOLS FORM TWO PRE-MOCK EXAMINATION
MARKING SCHEME
COMPUTER SCIENCE JUNE 2026

SECTION A (15 Marks)

Answer all questions in this section.

QUESTION 1 (10 Marks @1)

Each multiple-choice question carries 1 mark.

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	C	B	D	B	C	B	B	B	B

QUESTION 2 (5 Marks)

Each correct match carries 1 mark.

LIST A	(i)	(ii)	(iii)	(iv)	(v)
LIST B	B	E	D	C	A

Note: The list traces the five generations of computers in order — (i) Vacuum tube technology (1st generation), (iv) Transistor technology (2nd generation), (ii) Integrated circuits technology (3rd generation), (iii) Microprocessor design technology (4th generation), and (v) Very Large Scale Integration, aimed at artificial intelligence (5th generation).

SECTION B (67 Marks)

Answer ALL questions in this section.

3. (a) Analyze key infrastructure, communication systems, and industries that would completely collapse or require fundamental redesign in a world without computers. (5 marks)

Answer:

- (i) Banking and financial systems – electronic transactions, ATMs, and online banking would be impossible.
- (ii) Telecommunication networks – mobile networks, the internet, and digital communication would collapse.
- (iii) Transportation systems – air traffic control, GPS navigation, and modern logistics rely entirely on computers.
- (iv) Healthcare systems – electronic patient records, medical imaging, and diagnostic equipment depend on computers.
- (v) Government and administrative systems – digital record-keeping, tax systems, and public service delivery would need complete redesign.

Marking:

- 1 mark for each correctly identified infrastructure/industry, with a brief valid justification.

(b) Is there any link between one computer generation and the next one? Justify. (3 marks)

Answer: Yes, there is a link. Each computer generation builds upon the technological achievements and limitations of the previous one; for example, the invention of the transistor (2nd generation) replaced the bulky, unreliable vacuum tubes (1st generation), which in turn led to integrated circuits (3rd generation) and later the microprocessor (4th generation). Each generation therefore represents an improvement in speed, size, reliability, and cost based on the

discoveries of the generation before it.

Marking:

- 3 marks: A clear 'Yes' answer with valid justification showing progressive development between generations.
- 1-2 marks: A partial justification without clear linkage.

4. (a) Show how computers are useful in different areas like education, business and communication. (5 marks)

Answer:

- (i) Education – used for e-learning, research, and record keeping of students' performance.
- (ii) Business – used for accounting, online transactions, and stock/inventory management.
- (iii) Communication – used for sending emails, video conferencing, and instant messaging.
- (iv) Health – used for storing patient records and assisting in medical diagnosis.
- (v) Government – used for digital administration, planning, and record management.

Marking:

- 1 mark for each correctly identified area with a valid use.

(b) Why is it important for computer scientists to understand related disciplines? (5 marks)

Answer:

- (i) It helps them develop solutions that meet real needs in fields such as medicine, business, and engineering.
- (ii) It enables effective collaboration with professionals from other disciplines when designing systems.
- (iii) It improves their understanding of user requirements in different domains.
- (iv) It helps them apply mathematical, scientific and logical principles correctly in problem solving.
- (v) It enhances innovation by combining knowledge and ideas from different fields.

Marking:

- 1 mark for each correctly explained reason.

5. (a) Briefly explain why Jelvin wanted to understand the connection between computer science and information technology in real world life. (5 marks)

Answer:

- (i) To understand how computer science theory is applied practically through information technology.
- (ii) To see how software and hardware knowledge supports the use of technology in real life.
- (iii) To identify career opportunities that combine both computer science and IT.
- (iv) To understand how businesses and organizations use IT systems built on computer science foundations.
- (v) To appreciate how computer science concepts translate into everyday computing solutions and services.

Marking:

- 1 mark for each valid and relevant point.

(b) Jolyn noticed that computer science does not depend only on one field that contributes to its development. Identify four fields that contribute to computer science. (4 marks)

Answer: (i) Mathematics (ii) Electronics/Engineering (iii) Information Technology (iv) Statistics/Data Science

(Accept other valid fields like: Physics, Linguistics, Psychology/Cognitive science, Philosophy.)

Marking:

- 1 mark for each correctly named field. Total 4 marks.

6. (a) Identify five advantages of using the internet. (5 marks)

Answer:

- (i) Access to a wide range of information and research materials.
- (ii) Enables fast communication through email, social media and video calls.
- (iii) Supports online business and e-commerce.
- (iv) Facilitates e-learning and access to online education.
- (v) Provides entertainment such as streaming, music and games.

Marking:

- 1 mark for each correctly identified advantage.

(b) With examples, differentiate between system software and application software and explain how they interact to provide a functional experience for the user. (5 marks)

Answer:

- (i) System software manages and controls computer hardware and provides a platform on which other software can run, e.g. an Operating System such as Windows.
- (ii) Application software helps the user perform specific tasks, e.g. Microsoft Word for word processing.
- (iii) System software runs in the background to keep the computer functioning, while application software is opened directly by the user when needed.
- (iv) System software is essential for the computer to operate at all, while application software is optional and task-specific.
- (v) Interaction: The system software (operating system) provides the environment and manages hardware resources that the application software needs in order to run, allowing the user to interact with both hardware and applications smoothly.

Marking:

- 1 mark for each correct point of differentiation/explanation of interaction. Total 5 marks.

7. (a) Maria wants to perform preventive maintenance on a computer. What conditions should she check or ensure before starting the process? (5 marks)

Answer:

- (i) Ensure the computer is switched off and unplugged from the power source.
- (ii) Gather the necessary tools (screwdriver, soft brush, air blower, anti-static cloth).
- (iii) Work in a clean, dust-free and static-free environment.
- (iv) Back up important data before starting maintenance.
- (v) Ensure hands are clean and dry, and take anti-static precautions (e.g., wearing an anti-static wrist strap).

Marking:

- 1 mark for each correctly stated condition/precaution.

(b) What caution did you take for cleaning dust on hardware — computer mouse, monitor and keyboard? (5 marks)

Answer:

- (i) Switch off and unplug the computer before cleaning.
- (ii) Use a soft, dry cloth or soft brush to remove dust.
- (iii) Use an air blower to remove dust/debris from between keyboard keys.
- (iv) Avoid spraying liquid directly onto the screen, mouse or keyboard; apply cleaning solution to the cloth instead.
- (v) Avoid using excessive force or pressure when wiping the monitor screen.

Marking:

- 1 mark for each correctly stated precaution.

8. (a) Which computer parts might be affected if a computer is not properly maintained for a long time? (5 marks)

Answer: (i) CPU/motherboard (overheating due to dust build-up) (ii) Cooling fans (iii) Hard disk

drive (iv) Keyboard and mouse (dirt build-up causing malfunction) (v) Power supply unit

Marking:

- 1 mark for each correctly identified part.

(b) In what specific ways can malware harm a computer's performance, a user's private data, and the system's overall stability? (5 marks)

Answer:

- (i) Slows down system performance by consuming processing power and memory.
- (ii) Steals or exposes private/sensitive user data to unauthorized parties.
- (iii) Corrupts or deletes important files and programs.
- (iv) Causes frequent system crashes and instability.
- (v) Allows unauthorized remote access to the attacker, compromising the whole system.

Marking:

- 1 mark for each correctly stated harmful effect.

9. Define the following terms. (2 marks each — 10 marks)

(i) Cybersecurity

Answer: The practice of protecting computer systems, networks, programs and data from digital attacks, damage, or unauthorized access.

(ii) Cyber attack

Answer: A deliberate and malicious attempt by an individual or group to breach, exploit, or damage a computer system, network, or the data it holds.

(iii) Cyber threats

Answer: A potential danger or malicious act that could exploit a vulnerability to breach security and cause harm to a computer system or data.

(iv) Cybercrimes

Answer: Illegal activities carried out using computers, networks, or the internet, such as hacking, fraud, and identity theft.

(v) Hackers

Answer: Individuals who use technical skill and knowledge to gain unauthorized access to computer systems or networks, either with malicious intent or, in the case of ethical hackers, to identify and fix security weaknesses.

Marking:

- 2 marks for each correct and complete definition (1 mark for a partial definition). Total 10 marks.

SECTION C (15 Marks)

Answer question number ten (10).

10. Explain five components of cybersecurity. (15 marks)

Answer:

- (i) Network security: Protecting the computer network infrastructure from intruders, whether targeted attackers or opportunistic malware, using tools such as firewalls and intrusion detection systems.
- (ii) Application security: Keeping software and devices free of threats by building security into applications during design and continuously updating and testing them.
- (iii) Information/data security: Protecting the integrity, confidentiality and availability of data, both while it is stored and while it is being transferred.
- (iv) Operational security: The processes and decisions for handling and protecting data assets, including the permissions users have when accessing a network and the procedures for storing and sharing data.

- (v) End-user education: Training and educating users on safe computing practices, such as recognizing phishing emails, using strong passwords, and avoiding suspicious links, since human error is one of the biggest security risks.

Marking:

- 3 marks for each correctly named and explained component (1.5 marks for naming, 1.5 marks for explanation). Total 15 marks.
- Accept other valid components such as: disaster recovery/business continuity planning.